Case Study 1 - Rent vs. Buy

MECH 408 - Applied Engineering Economy
Colorado State University

Group 2
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Introduction:

This case study examines the financial implications of renting vs. buying a home for a PhD student moving to Fort Collins, Colorado, for a six-year graduate program at Colorado State University (CSU). When making long-term housing decisions, one of the largest financial dilemmas is to decide on buying or renting a home. This decision depends on various factors like rent increases, home appreciation, loan interest rates, and maintenance costs.

The goal of this analysis is to utilize Excel to aid in determining which option—renting vs. buying—is more ideal financially over the six years. This case study accounts for initial costs, monthly expenses, and potential long-term financial benefits such as property appreciation and mortgage principal reduction. Utilizing a detailed Excel model, we can calculate the Net Present Values (NPV) for both options, while considering factors like rent increases, mortgage payments, home value changes, and selling costs.

This case study includes testing for uncertainties like mortgage interest rate changes and property appreciation to assess their impacts on the data. After reviewing all of the data, this report will assess and create a recommendation on whether to buy or rent with the given criteria.

Methodology & Assumptions:

To conduct this case study we have taken the provided criteria and applied the following formulas to tabulate the NPV for each option. To determine the NPV of option 1 we have broken down the costs of renting on a monthly basis from 0 to 72 months (6 years) and calculated the present value for the sum of each month's costs. To determine present value (PV) we used the following formula:

$$PV = \frac{F}{(1+i)^n}$$

Where ‘F’ is the future value or total monthly cost, ‘i’ is the monthly discount rate (5% divided by 12), and ‘n’ is the month for which we are calculating the PV.

In option 1, at month 0, our only cost is the non-refundable deposit. For each subsequent month we total the cost of rent, utilities and insurance to find that respective month's total costs. Each year the rent is increased by 4% and so after every year we calculate the new rent using the future value equation:

$$F = P(1+i)^n$$

Where ‘F’ is the new value of rent, ‘P’ is the original rent value, ‘i’ is the annual rent increase rate, and ‘n’ is the previous year value.

Once the costs for every month are tabulated we then take the sum of each month’s PV from month 0 to 72 to determine the NPV for the entirety of option 1.

$$NPV = \sum_{n} PV$$

To determine the NPV for option 2 we use the same approach as option 1, where we calculate the costs for each month, determine the PV for that month, and then sum all of the month’s PV’s to find our NPV.

To determine the monthly cost first we must determine the mortgage amount. To do this we take the home price, subtract our down payment and add the closing costs:

$$Mortgage = Home\ Price - Down\ Payment + Closing\ Costs$$

The down payment and closing costs are 5% and 1% respectively of the home price, so we calculate those values by multiplying the home price by the respective percentages.

Now that we have a present mortgage value, we can calculate the monthly mortgage payment. In order to do this we use the NPV formula for a uniform series where we solve for the uniform amount:

$$A = P\left[\frac{i(1+i)^n}{(1+i)^n - 1}\right]$$

Where 'A' is the monthly mortgage cost, 'P' is the present mortgage value, 'i' is the monthly loan interest rate, and 'n' is the total number of months on the mortgage (number of years for the entire mortgage times 12 months/year).

The other monthly costs are either fixed values or are dependent on the home value. The monthly HOA is given and so we just use the cost provided. The annual maintenance is also given, so we break that down into a monthly cost by dividing by 12.

The property tax is dependent on the home value and is a given percentage of the home's worth, for this we assume the tax is reassessed every year and increases as the home appreciates in value. To calculate the monthly property tax we used the following formula:

$$PT = H(1+i)^{n-1} * \frac{r_t}{12}$$

Where 'PT' is the property tax cost, 'H' is the original home value, 'i' is the annual appreciation percentage value, 'n' is the current year for the given month, and 'r_t' is the annual tax rate.

At the end of year 6 we will sell the house so we have to calculate the selling cost/benefit of the house at this time. To determine the selling benefit we will take the future value of the home at this point, subtract our own closing costs, and also subtract the remaining principal value that will still be owed to the bank:

$$Selling\ Benefit = Future\ Value - Closing\ Costs - Princpal\ Remaining$$

To start we must determine the future value of the home. To do this we can use the future value (F) equation:

$$F = P(1+i)^n$$

Where 'P' is the original home value, 'i' is the annual appreciation rate, and 'n' is the number of years we intend to own the home. Next, we can calculate the selling costs with the provided closing cost rate. Assuming we sell the price for its full appreciated value, no more no less, we can multiply the future value we determined above by this percentage to determine our selling costs.

The last piece is to determine the principal amount still owed to the bank. To do this we created a loan payment table where for each month we take the present mortgage value, add the interest accrued based on the monthly mortgage loan rate, and then subtract the monthly mortgage payment which we calculated earlier (the 'A' value above). This gives the new monthly mortgage value for the next month, and so we can repeat this process for as many months as needed to work out our principal remaining for any given month.

Assuming we make our mortgage payment for the 72nd month and then sell the house before the 73rd month we can take the mortgage value for the 73rd month as our principal value that is still owed to the bank when we sell the house.

We can now calculate the selling benefit using the equation mentioned above. This selling benefit is added to the total costs/benefits of month 72 in our spreadsheet. With all costs/benefits now considered for option 2, we can calculate the PV for each month using the same formula mentioned above and then take the total of the PV's for all 72 months to determine option 2's NPV.

Results & Analysis:
Inputs:

- Discount Rate (Nominal): 5%
- Timeframe: 6 years (72 months)

Rent Option:	**Buy Option:**
● Monthly Rent: $1,400 ● Utilities: $250 ● Insurance: $30 ● Security Deposit: $1,200 ● No property tax, HOA, or mortgage costs	● Mortgage: $1,756.40/month ● HOA: $200 ● Maintenance: $250 ● Property Tax: $148.96 ● Insurance: $25 ● Closing Cost: $2,750 ● Initial Mortgage Loan: - $264,000 ● Home Sale Value (end of term): $238,000 (approx.)

Net Present Values (NPV's):

- **Rent Option NPV: -$114,140.02** (negative NPV, indicating financial cost)
- **Buy Option NPV: -$100,517.30** (negative NPV, indicating financial cost)

Summary:

Renting	**Buying**
Predictable monthly costs: $1,400 rent + $250 utilities + $30 insurance	Mortgage: ~$1,756/month
No large upfront investment	Additional costs: HOA, property tax, maintenance
No property tax, HOA, or maintenance responsibility	Requires an initial $2,750 closing cost and a $264,000 mortgage loan
Total cost over 6 years: $127,440 (excluding rent increases)	Future home sale value contributes to equity
NPV: -$114,140.02, suggesting long-term financial costs	**NPV**: -$100,517.30, suggesting long-term financial costs



Figure 1: Rent vs Buy Cumulative Cost-Months Excel

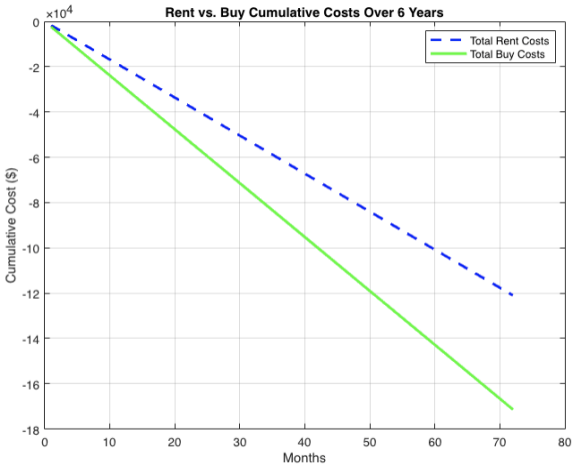


Figure 2: Rent vs Buy Cumulative Cost-Months Matlab

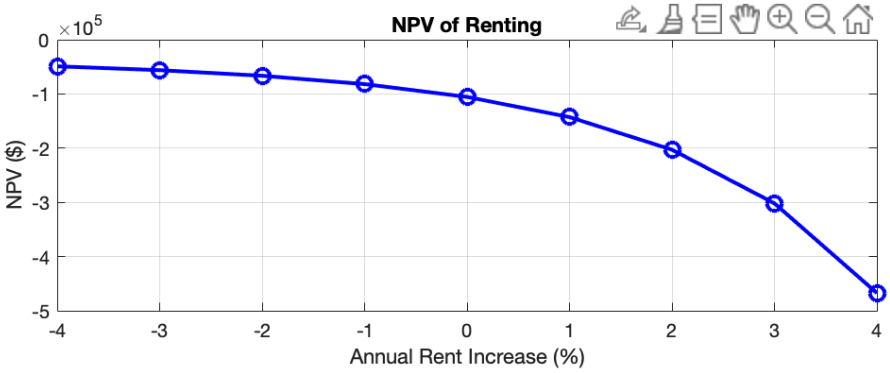


Figure 3: NPV of Renting-Annual Rent Increase

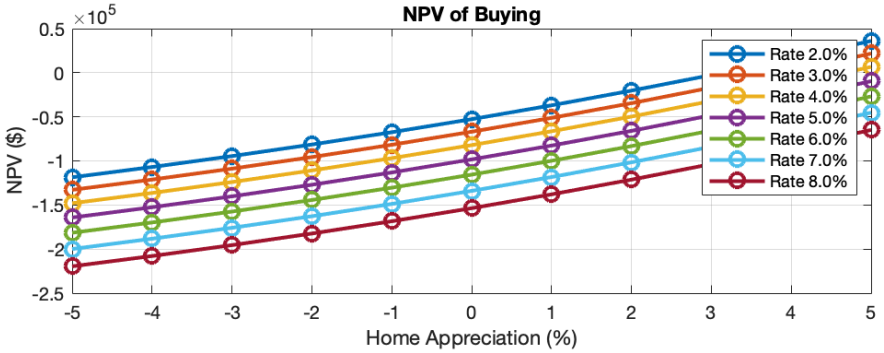


Figure 4: NPV of Buying-Home Appreciation Rate

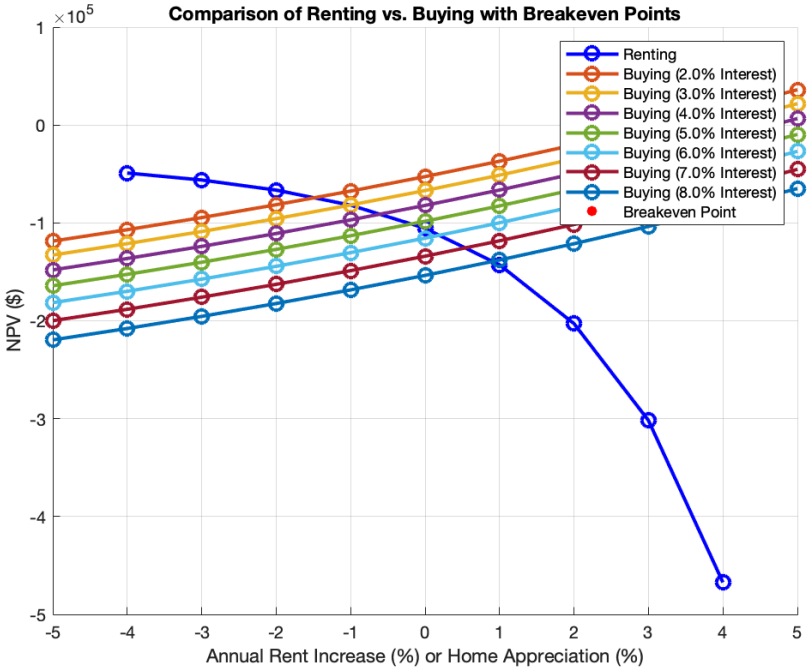


Figure 5: NPV of Renting vs Buying

Analysis & Discussion

As it can be seen in the graphs generated by the Excel and Matlab models, seen in *Figures 1* and *2* respectively, in terms of net costs to the graduate student in this scenario Renting the apartment yields fewer net costs at the end of the six year time period. However, when calculating the NPV of both options and then comparing the output result, we can make a more informed financial decision. This comparison can be seen in *Figure 5* above.

With our fixed timespan of 6 years it can be noted that in recessive economic conditions where the rate of change of property values and thus rents is low or negative, and where it can be expected for the interest rate to be set at a low percentage by the Federal Reserve, the NPV of the Rent option tends to

be higher. While in the opposite circumstance with rising property values, rents, and an interest rate which is set at a high level, the Buy option tends to have a superior NPV.

In less extreme economic conditions where the % rate of change of the value of property or rent is between -1% and 1% and the interest rate is 2-5% and 5-8% respectively, the decision maker can be relatively ambivalent to the difference in NPV as in these regions the variance is within 3.3kUSD.

To give a greater basis on which decision to make, a sensitivity analysis of both the rent and buy options was performed. Figure 6 below, shows the results of the sensitivity analysis for the NPV of option 1 and how the variables shown in *Table 1* can have an impact. As we can see the annual rent increase has the greatest impact on the NPV as it is the largest monthly cost.

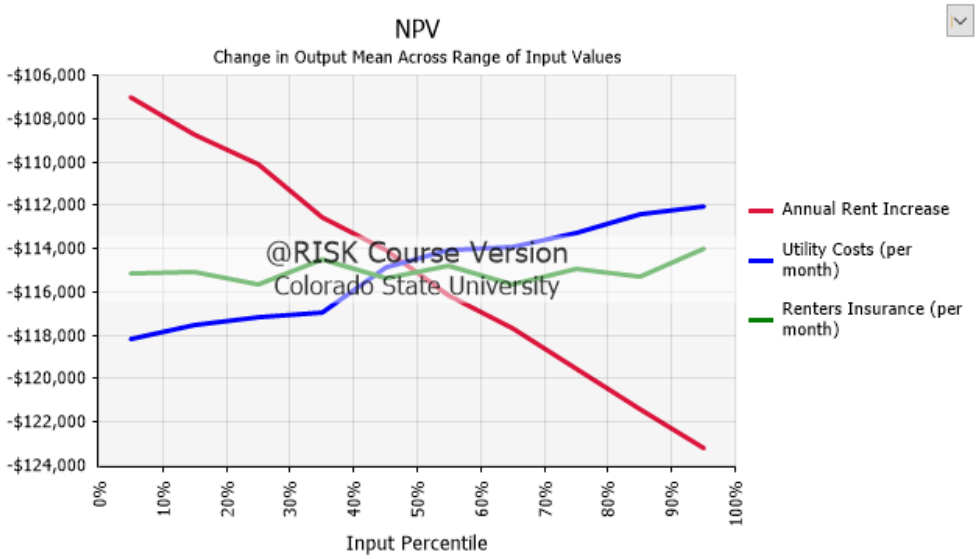


Figure 6: RISK Evaluation for Renting

Option 1: RISK Distribution Values		
Variable	Min Value	Max Value
Annual Rent Increase	0%	8%
Utilities	$200	$320
Renters Insurance	$27	$37

Table 1: RISK Distribution for Renting

Figure 7 below, shows the sensitivity analysis for the NPV for option 2 with *Table 2* giving the different variable bounds. As can be seen, only the variables which compound have a significant effect.

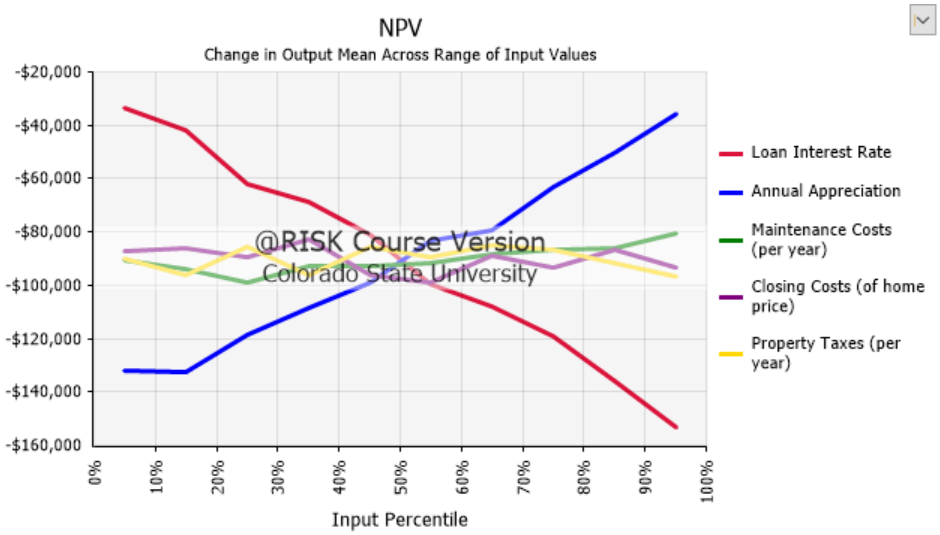


Figure 7: RISK Evaluation for Buying

Option 2: RISK Distribution Values		
Variable	Min Value	Max Value
Bank Loan Interest	2%	8%
Appreciation	-5%	5%
Property Taxes	0.50%	0.80%
Closing Costs	0%	3%
Maintenance	$2,200	$4,000

Table 2: RISK Distribution for Buying

Conclusion & Recommendation

Based on the analysis, both renting and buying have their merits and drawbacks. From a strictly financial perspective, buying tends to be the superior option overall.

When further analyzing the risk of this decision, purchasing the property for the six year period is still the better option. Looking at *Figures 6* and *7* two things should be noted. First, option 1 tends to have a lower NPV than option 2 no matter the variable intensity, and second, that the compounding effects have the greatest impact on the NPV of the respective decisions by far. The significance of the compounding effects should be given special attention, as whilst the option to buy has two separate sources of compounding, the property appreciation and interest rate, these two rates *should* approximately cancel one another out. In a faltering economy where property appreciation rates will be low, it can be expected that the interest rate will also be kept low by the Federal Reserve, whilst in the opposite scenario both should be high at the same time leading to the same effect. Compare this to the rent option, wherein there is only one source of compounding with no opposition, and thus if the economy is doing extremely well and thus property values and rents are increasing then the renter has no cushion whilst if the economy is doing poorly and thus rents are falling then the renter is doing extremely well. If one is seeking the lowest risk option available, then purchasing the property would provide that avenue.

However, financial considerations alone do not dictate housing choices. Renting remains a preferable option if flexibility and lower upfront costs are priorities. The ability to relocate without the burden of selling a property, freedom from maintenance responsibilities, and protection from market downturns make renting attractive, particularly for a short-term stay like six years. Furthermore, if mortgage rates increase beyond 6% or property values stagnate (or decline), the financial advantage of homeownership diminishes.

Ultimately, the decision should weigh both quantitative and qualitative factors. If long-term financial stability and investment potential are key considerations, buying is the recommended choice, provided market conditions are favorable. However, if uncertainty regarding future career plans, mobility, or market fluctuations exists, renting provides a lower-risk, flexible alternative. Given the variability in housing markets and mortgage rates, it is advisable for prospective buyers to closely monitor interest rate trends and real estate market conditions before committing to a purchase. Personal circumstances and risk tolerance should also play a critical role in the final decision, but for this particular study buying is preferred and recommended as shown on the in depth analysis.

Appendix

Matlab Codes~

Graph 2:

```
% Define variables
months = 1:72; % 6 years (72 months)

% Rent costs (cumulative)
monthly_rent = 1400;
utilities_rent = 250;
insurance_rent = 30;
total_rent_costs = cumsum((monthly_rent + utilities_rent + insurance_rent) * ones(1, 72));

% Buy costs (cumulative)
mortgage = 1756.40;
hoa = 200;
maintenance = 250;
property_tax = 148.96;
insurance_buy = 25;
total_buy_costs = cumsum((mortgage + hoa + maintenance + property_tax + insurance_buy) * ones(1, 72));

% Plot the costs over time
figure;
plot(months, total_rent_costs, 'b--', 'LineWidth', 2); hold on;
plot(months, total_buy_costs, 'g-', 'LineWidth', 2);
xlabel('Months');
ylabel('Cumulative Cost ($)');
title('Rent vs. Buy Cumulative Costs Over 6 Years');
legend('Total Rent Costs', 'Total Buy Costs');
grid on;
```

Graphs 3,4, & 5:

```
% MATLAB Script for Rent vs Buy Analysis

% Given Data Taken From Google Sheets Calcs
discount_rate = 0.05; % 5% nominal
timeframe_years = 6;
timeframe_months = 72;
initial_rent = 1400;
utilities = 250;
insurance = 30;
security_deposit = 1200;
initial_mortgage = 264000;
monthly_mortgage_payment = 1756.4;

% Sensitivity Ranges
interest_rates = linspace(0.02, 0.08, 7); % 2% to 8%
home_appreciation = linspace(-0.05, 0.05, 11); % -5% to +5%
rent_increase = linspace(-0.04, 0.04, 9); % -4% to +4%

% Preallocating Matrices for NPV Calculations
npv_rent = zeros(length(rent_increase), 1);
npv_buy = zeros(length(interest_rates), length(home_appreciation));

% NPV Calculation for Renting
for i = 1:length(rent_increase)
    rent = initial_rent;
    total_cost = security_deposit;
    for month = 1:timeframe_months
        total_cost = total_cost + (rent + utilities + insurance) / (1 + discount_rate/12)^month;
        rent = rent * (1 + rent_increase(i));
    end
    npv_rent(i) = total_cost;
end

% NPV Calculation for Buying
for i = 1:length(interest_rates)
    for j = 1:length(home_appreciation)
        mortgage = initial_mortgage;
        home_value = 275000; % Initial home value
        total_cost = 0;
```

```
% NPV Calculation for Buying
for i = 1:length(interest_rates)
    for j = 1:length(home_appreciation)
        mortgage = initial_mortgage;
        home_value = 275000; % Initial home value
        total_cost = 0;
        for month = 1:timeframe_months
            interest = mortgage * (interest_rates(i)/12);
            principal = monthly_mortgage_payment - interest;
            mortgage = mortgage - principal;
            home_value = home_value * (1 + home_appreciation(j)/12);
            total_cost = total_cost + (monthly_mortgage_payment * 200 + 250 + 25 + (home_value * 0.0065/12)) / (1 + discount_rate/12)^month;
        end
        selling_price = home_value * 0.94 - mortgage; % 6% selling cost
        npv_buy(i, j) = total_cost - selling_price;
    end
end

% Plot Results
figure;
subplot(2,1,1);
plot(rent_increase*100, npv_rent, 'b-o', 'LineWidth', 2, 'MarkerSize', 8);
xlabel('Annual Rent Increase (%)'); ylabel('NPV ($)'); title('NPV of Renting'); grid on;

subplot(2,1,2);
for i = 1:length(interest_rates)
    plot(home_appreciation*100, npv_buy(i, :), '-o', 'LineWidth', 2, 'MarkerSize', 8);
    hold on;
end
xlabel('Home Appreciation (%)'); ylabel('NPV ($)'); title('NPV of Buying'); grid on;
legend(arrayfun(@(x) sprintf('Rate %.1f%%', x*100), interest_rates, 'UniformOutput', false));
hold off;

% Combined Rent vs. Buy Graph with Breakeven Points
figure;
hold on;
plot(rent_increase*100, npv_rent, 'b-o', 'LineWidth', 2, 'MarkerSize', 8, 'DisplayName', 'Renting');
for i = 1:length(interest_rates)
    plot(home_appreciation*100, npv_buy(i, :), '-o', 'LineWidth', 2, 'MarkerSize', 8, 'DisplayName', sprintf('Buying (%.1f%% Interest)', interest
```

```
% Combined Rent vs. Buy Graph with Breakeven Points
figure;
hold on;
plot(rent_increase*100, npv_rent, 'b-o', 'LineWidth', 2, 'MarkerSize', 8, 'DisplayName', 'Renting');
for i = 1:length(interest_rates)
    plot(home_appreciation*100, npv_buy(i, :), '-o', 'LineWidth', 2, 'MarkerSize', 8, 'DisplayName', sprintf('Buying (%.1f%% Interest)', interes
end

% Find Breakeven Points
[x_breakeven, y_breakeven] = polyxpoly(rent_increase*100, npv_rent, home_appreciation*100, npv_buy(ceil(length(interest_rates)/2), :));
scatter(x_breakeven, y_breakeven, 100, 'r', 'filled', 'DisplayName', 'Breakeven Point');

xlabel('Annual Rent Increase (%) or Home Appreciation (%)');
ylabel('NPV ($)');
title('Comparison of Renting vs. Buying with Breakeven Points');
legend;
grid on;
hold off;
```